

2.5 PreQuiz: Motion, Lab Skills, and Spreadsheets

1. Round each value to three significant figures.

(a) 0.004817 m

(c) 7.0036 s

(b) 63.249 kg

(d) 21,560 N

2. Perform each operation and give the answer in scientific notation, rounded to 3 significant figures.

(a) $(3.2 \times 10^4) + (7.5 \times 10^3)$

(c) $\frac{6.3 \times 10^5}{9.0 \times 10^2}$

(b) $(8.40 \times 10^{-3}) \times (2.0 \times 10^4)$

3. Convert each value. Show one line of work using unit factors.

(a) 36.0 km/h to m/s

(b) 15.0 m/s to km/h

4. A student jogs along a straight track (forward is +). Start $x_0 = 5.0$ m.

(a) She jogs to $x = 17.0$ m. Find displacement Δx .

(b) From $x = 17.0$ m she turns back to $x = 9.0$ m. Find this Δx .

(c) Find total displacement from start to finish.

(d) Find total distance traveled.

Kinematics Equations

Formulas: $v = v_0 + at$ $d = v_0t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2ad$ $d = \frac{1}{2}(v_0 + v)t$

5. A bicycle accelerates uniformly from rest at 1.5 m/s^2 .

(a) What is its velocity after 4.0 s?

(b) How far does it travel during this time?

6. A ball is thrown vertically upward with an initial velocity of 12 m/s . Use $g = 10 \text{ m/s}^2$.

(a) How long does it take to reach its maximum height?

(b) What is the maximum height reached?

Lab and Technology Skills

In this section you will use the **Graphical Analysis** app at `graphicalanalysis.app`.

7. Open the Graphical Analysis app and enter the following data for a cart moving along a track. Time is in seconds; position is in meters.

Time (s)	Position (m)
0.0	0.20
0.5	0.24
1.0	0.36
1.5	0.56
2.0	0.84
2.5	1.20
3.0	1.64

- (a) Create a graph of Position vs. Time. Label the axes and give the graph a title.
- (b) Apply a **quadratic curve fit** to the data. Write the equation and the values of the fit parameters A , B , and C below.
- (c) Recall: for constant acceleration, $x = \frac{1}{2}at^2 + v_0t + x_0$. Using the value of A from your fit, calculate the acceleration a .
- (d) Save the .gambl file to the shared Google Drive folder. Include your name in the file name. You will take a **screenshot** of your graph with the curve fit displayed for the next problem.

8. Spreadsheet skills. Open a new spreadsheet (Google Sheets or Excel).

(a) **Paste** your Graphical Analysis screenshot into the spreadsheet.

(b) In a cell, type the acceleration you calculated: label it “Acceleration (m/s^2)”.

(c) In the next column, enter the cart mass $m = 0.390$ kg. Label it “Mass (kg)”.

(d) Write a **formula** in a third column to calculate the net force: $F = m \times a$. Label it “Force (N)”.

(e) What value does the spreadsheet calculate for the force? _____